

FINDINGS AND METHOD

ALIGNNN and the PINK Pipeline

A second architecture, the full 554,054-material screen,
and what changed when every claim was checked



PART ONE

Where the project stands

PINK reproduced, then extended – and what the extension is for

What PINK does, in one slide

PINK (Liu *et al.*, *J. Mater. Inf.* 2025) predicts **lattice thermal conductivity** κ_L straight from a CIF file, without solving the phonon Boltzmann transport equation.

It does that in two stages. A crystal graph network predicts the two elastic moduli; a closed-form physical model turns those into κ_L . The physics stage has no free parameters – it is the Slack model, not a fit.

Why this is worth doing

A full DFT + PBTE calculation of κ_L takes days per material. This takes about a sixth of a second. That is the difference between checking a handful of candidates and screening half a million.

The two stages:

Stage 1 – learned

CIF $\rightarrow$ graph network $\rightarrow$ bulk modulus K , shear modulus G



Stage 2 – physics

$K, G \rightarrow$ Poisson ratio $\rightarrow$ sound velocity, Debye temperature, Grüneisen $\gamma \rightarrow \kappa_L$

What this project added

A second architecture

ALIGNN, trained on the identical split, so the two can be compared without caveats. It beats the CGCNN ensemble on both moduli.

The full screen, twice

All 33,118 GNoME candidates scored by both architectures – the paper's own headline experiment, run independently.

Validation against measurement

45 materials with experimentally measured κ_L , not model-vs-model.

And – the part that took longest –

every headline claim re-checked against the artefacts that produced it. Four evaluation failure modes were found, each of which inflates reported performance. Three numbers already written down turned out to be wrong.

Parts 6 and 7 of this deck are those checks. They are the least glamorous material here and probably the most useful to anyone else building the same kind of screen.

PART TWO

ALIGNN: from CIF to two graphs

The one structural idea that separates it from CGCNN

Why a second architecture at all

A single model that scores well can be right for the wrong reason. Two models that share only the training data and the physics – not the architecture – give an independent check: where they agree, the signal is unlikely to be an artefact of either one.

ALIGNN was chosen because its difference from CGCNN is **physically motivated rather than incidental**. It is not a bigger CGCNN or a differently tuned one. It sees something CGCNN structurally cannot: **bond angles**.

What CGCNN encodes

Which atoms exist, which are bonded, and *how far apart* they are.
A pure two-body description.

What ALIGNN adds

The *angles* between bonds – a three-body term. Elastic response to shear depends on how bonds resist *bending*, not only stretching.

Shear modulus is a bond-bending property. A model blind to angles is being asked to infer it indirectly.

Step 1 --- the CIF, and finding neighbours

Identical to CGCNN's start: a CIF gives lattice vectors, fractional coordinates and the element at each site. Neighbours come from a **periodic distance search**, so an atom near a face still finds those across the boundary.

The two settings that define the graph

`cutoff = 5.0 Å` – the radius within which a neighbour counts

`max_neighbors = 12` – the cap per atom

Read from the trained model's own config – building 8.0 Å graphs for a 5.0 Å model would feed it an unseen neighbourhood.

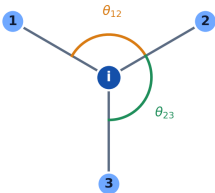
What comes out of this step:

quantity	becomes
element at site i	node feature $\mathbf{v}_i$
bond $i \rightarrow j$	edge $\mathbf{e}_{ij}$
distance r_{ij}	edge feature
angle θ_{jik}	ALIGNN only

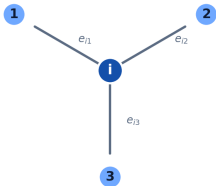
Node features come from the same 92-dimensional element table CGCNN uses (`atom_features = cgcnn` in the config), so the two architectures start from identical chemistry.

Step 2 --- one crystal becomes *two* graphs

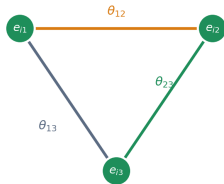
A the crystal
atoms, bonds, and the angles between them



B the ATOM graph $\mathcal{G}$
nodes = atoms, edges = bonds (CGCNN stops here)



C the LINE graph $\mathcal{L}$
nodes = bonds, edges = ANGLES (ALIGNN adds this)



A the real motif: a centre atom, its neighbours, and the angles between the bonds. **B** the **atom graph** $\mathcal{G}$ – atoms are nodes, bonds are edges. This is exactly what CGCNN builds, and where CGCNN stops. **C** the **line graph** $\mathcal{L}$ – every *bond* of $\mathcal{G}$ becomes a *node*, and two bonds sharing an atom are joined by an edge that carries the angle between them.

The line graph is a graph of the bonds. Its edges are angles – which is how a message-passing network gets access to three-body geometry.

Why the line graph is the right construction

A message-passing network updates a node from its neighbours. On the atom graph alone, the only geometric quantity available to a message is the bond *length* r_{ij} – there is nowhere for an angle to live, because an angle is a property of a *pair* of bonds, not of one.

Promoting bonds to nodes solves that exactly. In $\mathcal{L}$, a pair of bonds is an *edge*, and an edge is precisely where a feature can be attached.

Counting the geometry

An atom with n neighbours contributes n nodes to $\mathcal{L}$ and $\binom{n}{2}$ angle edges. At `max_neighbors = 12` that is up to 66 angles per atom – the reason ALIGNN

Angle encoding:

The angle is fed as $\cos \theta_{jik}$ expanded on a radial basis, the same way distances are. Using the cosine rather than the angle keeps the feature smooth where it matters and avoids a discontinuity at 0 and π .

A sanity check

A cubic cell and a sheared one with identical bond lengths are *indistinguishable* to a two-body descriptor, and distinct to ALIGNN.

Step 3 --- the ALIGNN layer alternates between them

One ALIGNN layer is two coupled updates, run in order:

1. Update on the line graph $\mathcal{L}$

Each bond-node is updated from the bonds it shares an atom with, using the *angle* on the connecting edge. Bond features now carry angular information.



2. Update on the atom graph $\mathcal{G}$

Each atom-node is updated from its neighbours using those *angle-aware* bond features as edge inputs.

Then, identically to CGCNN

Pool the atom-node vectors over the cell (mean), pass through a small fully-connected head, and emit one number.

Trained on $\log_{10}$ of the modulus, so the loss is relative – a 10 GPa error on a 20 GPa crystal is penalised far more than on a 300 GPa one.

Two separate models, one per target – as in the CGCNN arm.

CGCNN and ALIGNN, side by side

	CGCNN	ALIGNN
graphs built	atom graph only	atom graph + line graph
geometry seen	bond lengths (two-body)	lengths + angles (three-body)
node features	92-dim element table	same 92-dim table
neighbour rule	cutoff + max-neighbour cap	same (5.0 Å, 12)
message passing	on $\mathcal{G}$	alternating $\mathcal{L}$ then $\mathcal{G}$
cost per structure	lower	higher (up to $\binom{12}{2}$ angles/atom)
bulk K MAE $\log_{10}$	0.0630 (3-model ens.)	0.0539
shear G MAE $\log_{10}$	0.0781 (3-model ens.)	0.0725

Same 10,987 crystals, same `split_seed` = 42, same 1,648 held-out test set, both trained the full 150 epochs with no early stop. The comparison is architecture and nothing else.

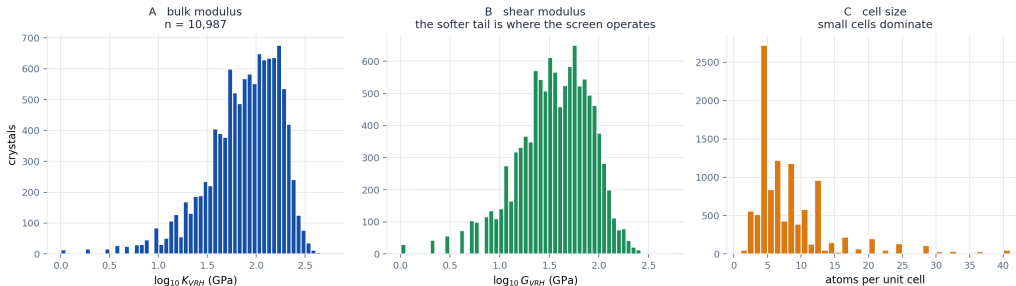
ALIGNN wins on both moduli – and it is the only model improvement in this project that survived the honest re-scoring in Part 7.

PART THREE

Training and accuracy

The dataset, the fit, and where it fails

The training set --- PINK's Figure 4, our version

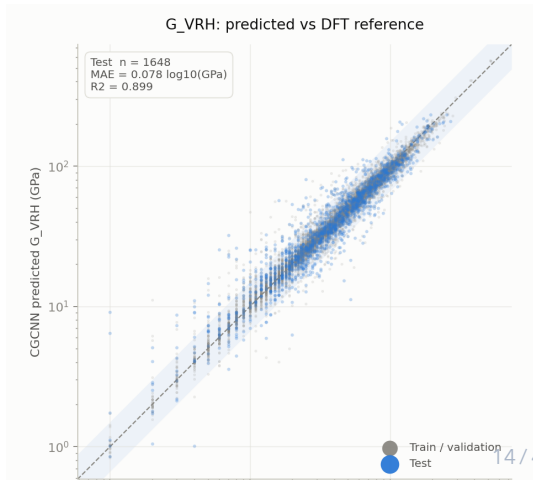
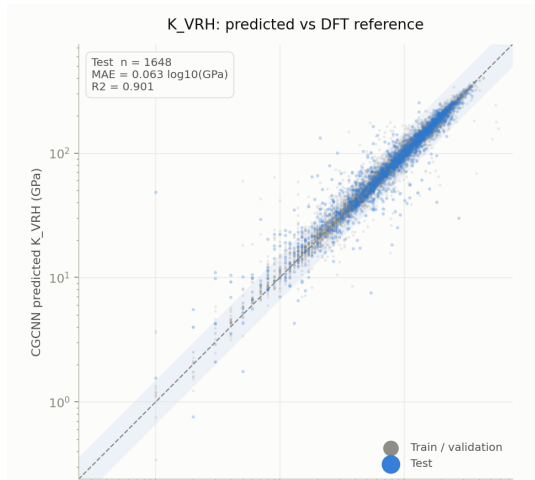


The matbench elastic benchmark: **10,987 crystals** with DFT-computed K_{VRH} and G_{VRH} , split 70/15/15 at `split_seed = 42` into 7,691 train / 1,648 validation / 1,648 test. Both moduli are strongly right-skewed, which is why everything here is trained on $\log_{10}$.

The soft tail in panel B is thin – and it is exactly the region the whole screen is aimed at. That imbalance shows up again in Part 3 and Part

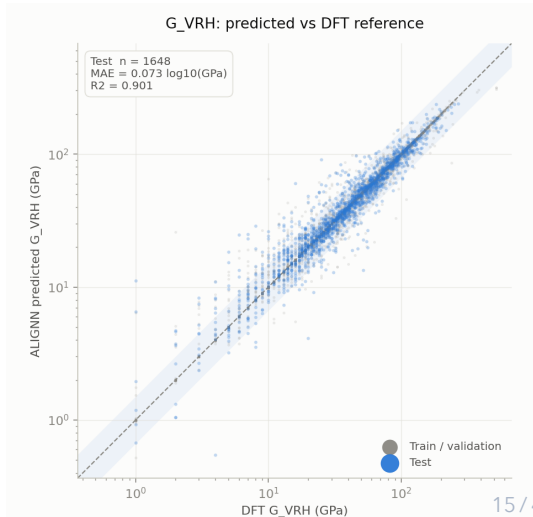
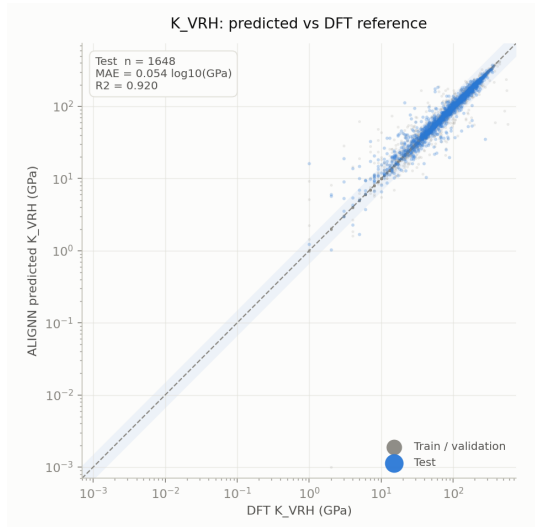
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Predicted vs DFT --- PINK's Figure 5, CGCNN arm



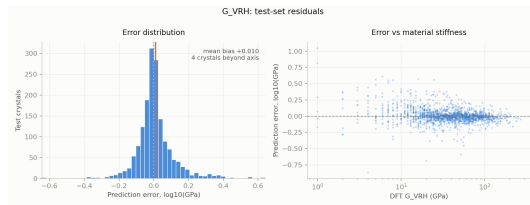
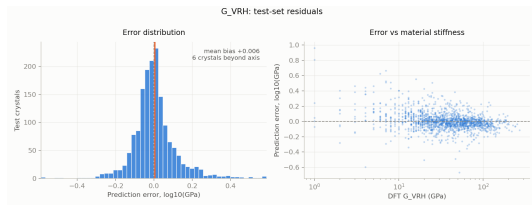
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Predicted vs DFT --- the ALIGNN arm



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Where the models fail --- residuals



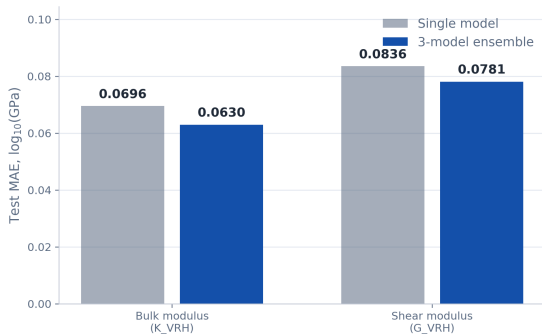
CGCNN ensemble, shear

ALIGNN, shear

Residuals fan out below ~ 30 GPa in both. Stratified by true κ_L , the softest quartile has MAE $\log_{10}(G) = 0.134$ against 0.057 for the stiffest – a factor of 2.4.

Both architectures are worst in exactly the regime the screen targets.
Part 7 asks whether that is the model or the labels.

What ensembling buys



Three models per target, different random initialisations, averaged in $\log_{10}$ space – a geometric mean, which is the right notion of middle for a quantity whose error is proportional.

Ensembling removed 9-10% of the error:
bulk $0.0696 \rightarrow 0.0630$, shear
 $0.0836 \rightarrow 0.0781$.

And a free uncertainty estimate

The spread across members gives a per-crystal error bar – the thing PINK's single pretrained model cannot produce at all.

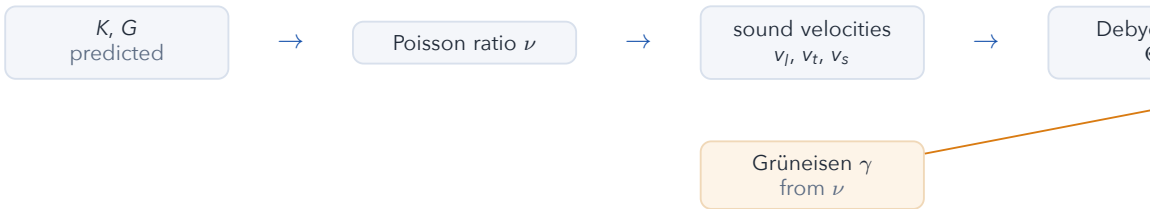
Part 4 shows that estimate is badly overconfident.

PART FOUR

The physics

From two moduli to a thermal conductivity

The Slack chain --- no free parameters

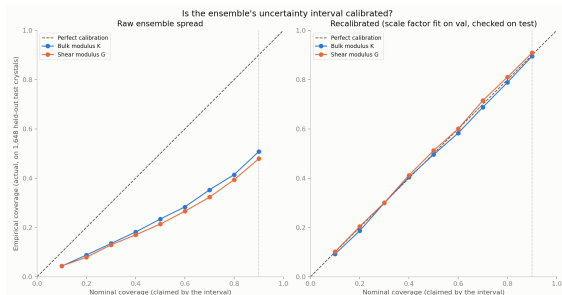


Every arrow is a closed-form relation. The only learned quantities in the whole chain are K and G ; everything downstream is algebra plus the structural constants (volume, density, atom count) read straight from the CIF.

γ is **derived** from the predicted K/G ratio through an empirical Poisson relation – matbench contains no measured γ at all. That single choice turns out to dominate several later results.

Because the physics is shared and unchanged, any difference between two of this project's screens is the *moduli*, never the formula.

The uncertainty interval was badly overconfident



The Monte Carlo p05-p95 band was checked against the test split's real DFT labels – no new data needed.

Claimed 90% coverage. Actual 48-51%.

Cause: three identically-trained members share blind spots, so member *disagreement* runs 4-6× smaller than true error.

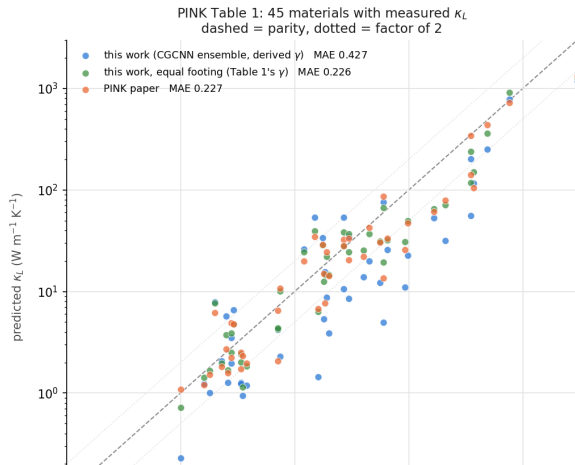
Recalibration factors from validation ($K \ 4.27\times$, $G \ 4.54\times$) recover $\sim 90\%$ coverage out of sample – and collapse almost every “tight” label on the DFT shortlist to “wide”.

PART FIVE

Validation against measurement

45 materials where the answer is known

Measured κ_L --- PINK's Figure 6B, our version



The paper's Table 1 lists 46 materials with **experimentally measured** κ_L . 45 survive (mp-3490/GaP is deprecated in Materials Project).

The range spans four orders of magnitude – AgCl at 1.0 to diamond at $3000 \text{ W m}^{-1} \text{ K}^{-1}$ – so everything is scored in $\log_{10}$.

mean $|\Delta \log_{10}|$

ours, derived γ 0.427

ours, equal footing **0.226**

the PINK paper 0.227

The comparison was never like-for-like

Table 1's γ values come from **AFLOW / experimental lookups** – the paper's own text says so. They are *not* produced by the Poisson-ratio formula that our pipeline uses.

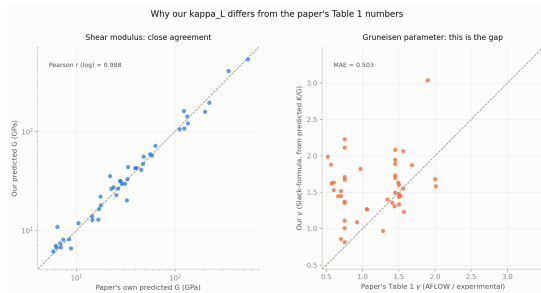
Neither does the paper's own released screening code use them: it uses the same formula we do. So comparing our derived γ against a looked-up- γ table tests **the γ source**, not the model.

Substituting Table 1's own γ

changes nothing else – same moduli, same physics –

0.427 → **0.226**

against the paper's 0.227.



Our moduli were never the problem: shear modulus and sound velocity agree with the paper's own tabulated values at $r \approx 0.99$.

The comparison sheet

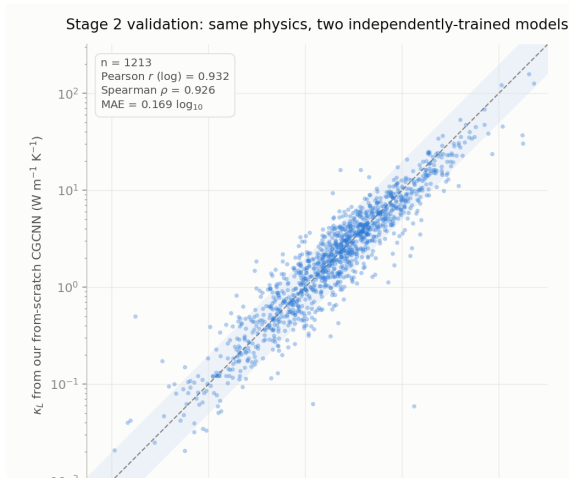
material	measured	ours (derived γ)	ours (equal footing)	PINK
AgCl	1.0	0.23	0.71	1.09
Bi ₂ Te ₃	1.6	1.24	1.43	1.20
NaI	1.8	1.00	1.66	1.52
PbTe	2.5	5.72	3.74	2.71
... 37 more ...				
AlN	350.0	55.87	118.49	140.93
SiC	490.0	251.19	360.10	438.31
BN	760.0	779.67	913.63	727.98
C (diamond)	3000.0	1180.18	1260.63	1320.87

results/cgcnn/63_experimental_vs_predicted.csv – 45 rows, 27 columns. Both γ conventions are carried per material, along with each model's moduli, all three γ values, and the p05/p95 interval, so the choice is visible rather than baked in.

ALIGNN is worse here – 0.450 against the CGCNN ensemble's 0.427.

It wins on matbench moduli and loses on measured κ_L . That was tested, not assumed.

Two architectures, independently, on the same 1,213 crystals



ALIGNN's moduli pushed through the *unchanged* Slack chain, compared against the CGCNN ensemble's κ_L on the same crystals.

Pearson $r = 0.940$, **Spearman** $\rho = 0.947$,
MAE $0.141 \log_{10}$ – tighter than CGCNN's
own agreement with the paper's pretrained
model ($r = 0.932$).

Two independently trained architectures, sharing only the data, the split and the physics, converging this closely on final κ_L is evidence the pipeline finds a physical signal rather than architecture-

PART SIX

The screen

554,054 materials, scored twice

The screening funnel --- PINK's Figure 7

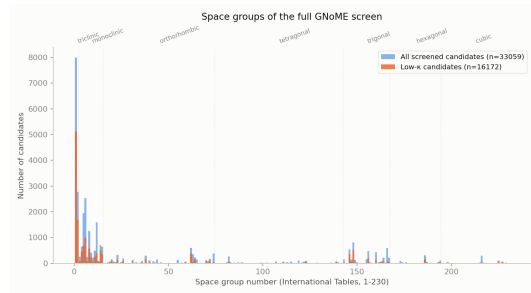
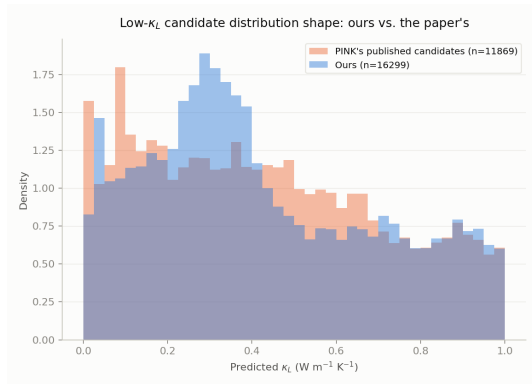


The current GNoME snapshot holds **554,054** materials against the paper's 377,221 – stated explicitly rather than glossed, because it means the candidate lists are not directly comparable.

Against the paper's own published 11,869 candidates, **70.8%** also appear in our independently derived list. Two pipelines sharing only an architecture family and a physics formula, on overlapping-but-different corpora, converging on most of the same discoveries.

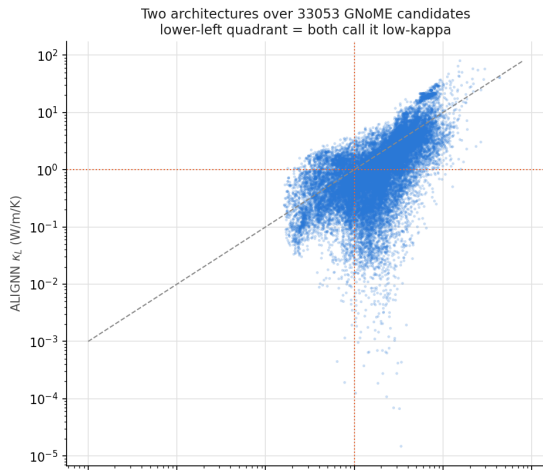
Flagged: that **70.8%** predates several corrections in Part 7 and should be recomputed before it is quoted in a paper.

What the candidates look like --- PINK's Figure 8



κ_L distribution and crystal-system breakdown of the screened pool. Element enrichment tells the clearest physical story: halogens $1.4\text{--}1.9\times$ enriched among low- κ candidates, oxygen $0.27\times$ depleted – soft heavy halide lattices against tight oxide bonding.

ALIGNN over the whole screen, for the first time



All 33,118 candidates scored with ALIGNN – 68 minutes, zero CIF failures. Only the moduli change; the Slack chain and the structural constants are identical, so nothing downstream can leak a difference.

low- κ call on 33,053 reliable candidates

ALIGNN 14,711 (44.5%)

round 9 7,262 (22.0%)

both agree 6,006

Spearman ρ 0.723

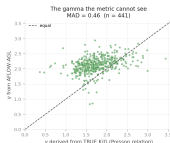
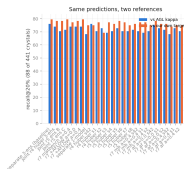
65 candidates hit ALIGNN's $\sim 10^{-3}$ GPa regression floor – a degenerate output that reads as a spectacular discovery. Flagged and

PART SEVEN

What we found when we checked

Four evaluation failure modes, and three numbers that were wrong

The screening metric was circular



recall@10% – the fraction of the true lowest- κ decile recovered – was scored against a reference that was *this project's* own Slack chain on matbench's true moduli, with γ derived by the same relation the prediction side uses.

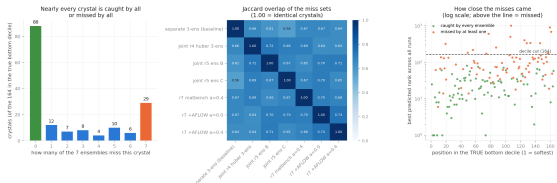
Slack cancels. γ cancels harder: **a model predicting a better γ scored worse.**

Re-scored against AFLOW-AGL's independent κ_L (441 matched crystals): recall@10% fell **20.5 points**, Spearman ρ fell **0.139**.

The old metric actively penalised the one remaining lever. **Quote** Spearman ρ – it is what transfers between targets ($\rho = 0.92$ across 26 runs); recall@10% does not ($\rho = 0.20$).

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The misses are a fixed core, not variance



Fingerprinting the saved prediction vectors collapsed 34 roster entries to **26 distinct** runs – eight were byte-identical duplicates.

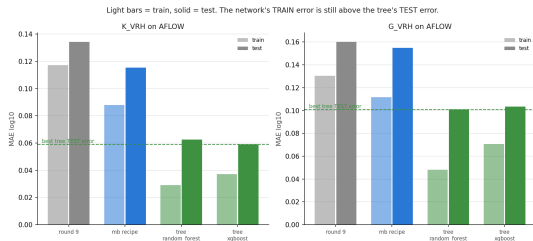
Across those 26, recall spans **64.0-70.7%**. Which crystals are missed is remarkably stable: **16 missed by all 26 runs**, 29 by all seven ensembles, 65 never missed.

The 16 permanent misses: TmAg_3 , MgAgAs , CeTe , $\text{Cs}_2\text{NaAlH}_6$, InSb , YSe and ten others – median 4 sites, 9 of 16 cubic, never ranked better than 173rd by anything.

The ceiling is in the inputs, not the architecture. Seven rounds of architecture work never moved it.

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A composition-only tree beats the network



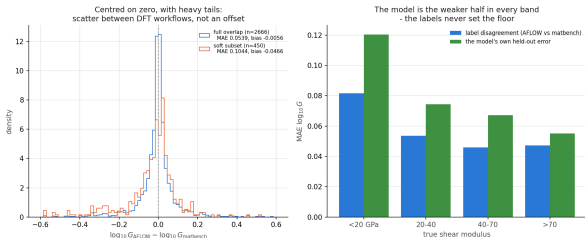
A decision tree on the composition-weighted mean of the same 92-dim element table – **no structure, no bonds** – on the identical split:

K MAE log ₁₀	tree	CGCNN
matbench	0.0868	0.0696
AFLOW	0.0592	0.1154

The cause is underfitting, checked directly.

Round 9's *training* error on AFLOW K is 0.103 – worse than the tree's *test* error of 0.059. A model that cannot fit its own training data is not out-generalised.

The label noise everyone quotes was the wrong number



A single constant – $|\Delta \log_{10} G| = 0.1085$ – was hard-coded into four scripts as *the* matbench/AFLOW label noise, and divided by a whole-test-set model error to argue the labels set the accuracy floor.

0.1085 is the *soft subset's* number.

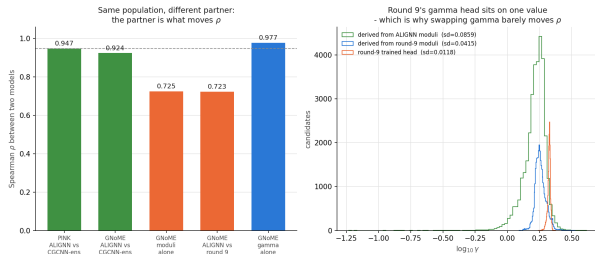
Full overlap (n = 2,666): **0.0539**

Population-matched ratio: 0.69, not 1.39

Per stiffness band the *model's* error exceeds the label disagreement in **all four**.

Also: the soft subset's apparent offset is a

Where a cross-model disagreement actually lives



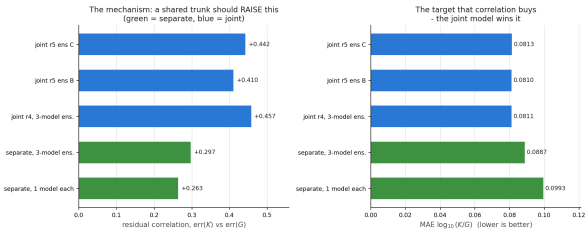
Agreement between ALIGNN and the CGCNN ensemble is $\rho = 0.947$ on the PINK set but only 0.723 against round 9 on GNoME. Two explanations were plausible: the exotic GNoME chemistry, or the partner model.

population cost (PINK $\rightarrow$ GNoME)	-0.023
partner cost (ens. $\rightarrow$ round 9)	-0.201

moduli alone	0.725
γ alone	0.977

Holding the partner fixed, GNoME's chemistry costs almost nothing. Swapping the partner costs nine times more – and it is entirely the moduli.

The joint model worked better than the record said



κ_L depends on K and G only through their *ratio*, and in log space a ratio is a difference – so independent errors **add**. A shared trunk should make them correlate instead.

	corr	MAE log ₁₀ (K/G)
separate, 1 each	+0.263	0.0993
separate, 3-ens	+0.297	0.0887
joint 3-ens	+0.457	0.0811

A published table had labelled the first two columns “separate” and “joint” – so it contained no joint model at all. The real correlation is 1.5× the ensemble’s.

PART EIGHT

What happens next

The one experiment that closes the gap

The missing piece --- and the experiment for it

Every result in this project is model-vs-model or model-vs-database. **No candidate has ever been checked against a calculation we ran ourselves.** That is the single thing blocking any discovery claim.

Arm A – consensus low

All four models agree. Ranked by the *most pessimistic* of them, not the lowest prediction.

8 of 69 eligible

Arm B – adjudication

Three models say low, round 9 says high. GNoME has no ground truth; DFT settles it.

6 of 214 eligible

Arm C – negative control

All four call it stiff. Without this, a confirmed Arm A means little.

4 of 297 eligible

Compute the elastic tensor, not κ . Two independent diagnostics localise every model disagreement here to the moduli, and a finite-strain elastic tensor on a 6–10 atom cell is a handful of static calculations. Quasi-harmonic Debye κ then follows from the same tensor almost free.

Start with MoBr_5Cl – 7 atoms, Cm , and the sharpest disagreement in the whole screen: ALIGNN 0.028 against round 9's 4.524, a factor of 162.

Where this can be published

The honest framing is a **reproducibility and evaluation paper**, not a materials-discovery paper. The contribution is a set of measured failure modes that inflate reported performance in this problem class:

1. a circular screening metric
2. uncalibrated ensemble intervals
3. a composition baseline beating an underfit GNN
4. population-dependent label noise

What blocks the discovery framing

No external validation – fixed by Arm B above.

No net improvement: seven rounds produced nothing that survived honest evaluation.

Small samples on key diagnostics ($n = 441$, $n = 43$) – state them.

Re-verify every headline number first. This project's own record contained three that were wrong.

IN ONE SLIDE

What this work actually established

The pipeline finds a physical signal. Two architectures sharing only data and physics agree at $\rho = 0.947$ on κ_L .

ALIGNN earns its extra structure. Bond angles via the line graph improve both moduli – the only model change that survived re-scoring.

On equal footing we match the paper against measured κ_L : 0.226 against 0.227.

Most reported gains here are measurement artefacts: a circular metric, an over-confident interval, an underfit network flattered by comparison, a label-noise constant from the wrong population.